

Assignment 1: Data Mining Practice and Theory

Group 62

Anna De Martin (2906611), Alicja Dorobis (2746727), and David Žantovský
(12609536)

Vrije Universiteit, Amsterdam 1081 HJ, Netherlands

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1 Introduction

This report aims to present an application of the CRISP-DM process model to a longitudinal smartphone sensor dataset collected to support research on mood changes and depression monitoring [1]. The dataset contains timestamped self-reports and passive sensor recordings from 27 participants over approximately four months. The primary objective is to predict a participant's mood the following day, framed as both a classification and a regression problem. The report covers exploratory data analysis, data cleaning and imputation, temporal feature engineering, classification using XGBoost and a Long Short-Term Memory (LSTM) network, numerical prediction using the same models, and an analysis of evaluation metrics.

2 Data Preparation

2.1 Exploratory Data Analysis

The raw dataset contains 376,912 rows in long format, where each row represents one timestamped reading of a single variable for one participant. The dataset spans February to June 2014 and covers 19 variables across two main categories: discrete ordinal self-report variables (mood on a 1-10 scale, circumplex arousal and valence each on a -2 to +2 scale) and passive sensor data (continuous duration variables from screen and apps grouped into 12 categories, activity score, and binary call and SMS event markers).

Table 1 summarizes key descriptive statistics per variable. From this first overview we notice that, first, variables are sampled at different frequencies, e.g. mood has 5,641 reading counts while screen time has 96,578. Second, mood is highly concentrated around 6-8 (mean = 6.99, std = 1.03), with fewer than 5% readings below 5, indicating strong right-skewness in its distribution (Fig 2). Third, circumplex valence shows a mean of 0.69, while arousal of -0.10. Valence represents the degree of pleasure - positive vs. negative (here generally positive), while arousal represents the level of physiological activation - high energy vs. calm (here almost neutral). These two variables provide a multi-dimensional view of affect that a single 1-10 mood score might miss, as highlighted by [2].

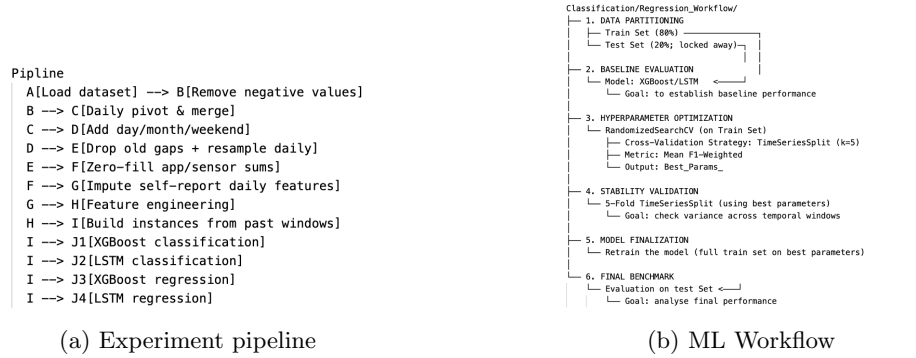


Fig. 1: Pipeline architecture

Table 1: Descriptive statistics of the raw longitudinal dataset variables.

Variable	Count	Mean	Std	Min	25%	Median	75%	Max	Missing	Missing %
activity	22,965	0.12	0.19	0.00	0.00	0.02	0.16	1.00	0	0.00
appCat.builtin	91,288	18.54	415.99	-82,798.87	2.02	4.04	9.92	33,960.25	0	0.00
appCat.communication	74,276	43.34	128.91	0.01	5.22	16.23	45.48	9,830.78	0	0.00
appCat.entertainment	27,125	37.58	262.96	-0.01	1.33	3.39	14.92	32,148.68	0	0.00
appCat.finance	939	21.76	39.22	0.13	4.07	8.03	20.16	355.51	0	0.00
appCat.game	813	128.39	327.15	1.00	14.15	43.17	123.62	5,491.79	0	0.00
appCat.office	5,642	22.58	449.60	0.00	2.00	3.11	8.04	32,708.82	0	0.00
appCat.other	7,650	25.81	112.78	0.01	7.02	10.03	16.83	3,892.04	0	0.00
appCat.social	19,145	72.40	261.55	0.09	9.03	28.47	75.37	30,000.91	0	0.00
appCat.travel	2,846	45.73	246.11	0.08	5.09	18.14	47.23	10,452.61	0	0.00
appCat.unknown	939	45.55	119.40	0.11	5.02	17.19	44.43	2,239.94	0	0.00
appCat.utilities	2,487	18.54	60.96	0.25	3.16	8.03	19.33	1,802.65	0	0.00
appCat.weather	255	20.15	24.94	1.00	8.68	15.12	25.35	344.86	0	0.00
call	5,239	1.00	0.00	1.00	1.00	1.00	1.00	1.00	0	0.00
circumplex.arousal	5,597	-0.10	1.05	-2.00	-1.00	0.00	1.00	2.00	46	0.80
circumplex.valence	5,487	0.69	0.67	-2.00	0.00	1.00	1.00	2.00	156	2.80
mood	5,641	6.99	1.03	1.00	7.00	7.00	8.00	10.00	0	0.00
screen	96,578	75.34	253.82	0.04	5.32	20.04	62.54	9,867.01	0	0.00
sms	1,798	1.00	0.00	1.00	1.00	1.00	1.00	1.00	0	0.00

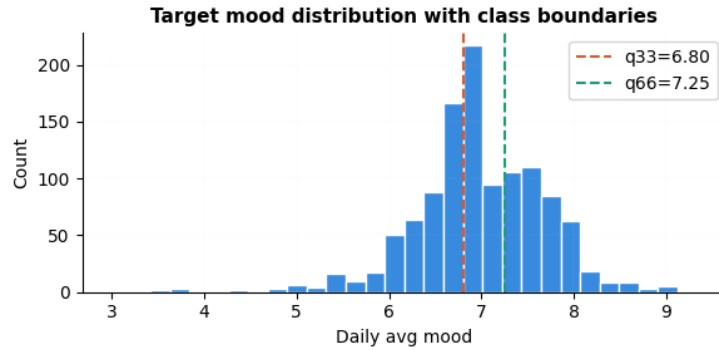


Fig. 2: Quantile distribution of mood

Fourth, duration variables have extreme outliers: screen time reaches a maximum of 9,867 seconds per reading (~ 2.7 hours), and appCat.social reaches 30,001 seconds (~ 8 hours) in a single recording.

We identified two other data quality issues. The variable appCat.builtin contains three readings with negative values (minimum = - 82,799 seconds), possibly a sensor recording errors. appCat.entertainment similarly contains a small negative value (- 0.011), likely attributable to floating-point noise. Since they are both duration measures, we consider their negative values as data errors. Conversely, analysing missing values in the Circumplex dimensions, we noticed data were missing at the observation level rather than the day level. In all but two cases, users had provided other inputs on the same day, suggesting that the missingness was a result of user behaviour (incomplete surveys) rather than hardware malfunctions. Consequently, we treated these as valid participant-days suitable for imputation, rather than erroneous data requiring deletion.

2.2 Data Cleaning

Outlier and Error Removal We removed the three negative rows in appCat.builtin and the single negative row in appCat.entertainment, as they represented physically impossible negative duration values. No further outlier removal was applied to duration variables: while extreme values exist (e.g. 9,867 seconds of screen time in a single reading), these may legitimately reflect heavy smartphone-usage days, and applying any sort of clipping could remove genuine variation relevant to mood prediction.

Imputation Strategies Before imputation, we pivoted the long-format dataset to wide format by aggregating each variable per participant per calendar day. Self-report variables (mood, arousal, valence) and activity score were aggregated using their daily mean value. Maximum, minimum, standard deviation and count values were also recorded for analysis purposes. Duration (screen and applications) and event variables (calls and sms) were summed to give total daily usage. This resulted in a wide-format dataset of 1,973 participant-day rows and 40 columns before imputation.

We also identified participants with significant gaps in their mood recordings. Specifically, three individuals did not record their mood state for periods lasting 7, 12, and 22 days. We decided to remove all data collected before these large gaps, keeping only the most recent continuous block of observations for each person. This was done to avoid structural breaks: after longer periods of time, a participant's life circumstances or mental state could have shifted so much that the older data would only add noise or bias to our model. By focusing on the most recent uninterrupted data, we ensure our features stay relevant. After this cleanup, the final dataset contained 1,268 participant-day rows.

For imputation, application category variables were zero-filled rather than imputed, as a missing value in this context means the application category was not used on that day. The same logic applies to screen, call and sms.

For the self-report variables (mood, arousal, valence) and activity score, we evaluated two distinct imputation strategies to handle temporal gaps of up to three days: Last Observation Carried Forward (LOCF), which assumes a participant’s psychological state remains static until a new report is logged, and Linear Interpolation, which assumes a constant rate of change between two known observations, creating a smoother transition. Our decision was informed by [3] and [4]. In particular, the latter highlights that while LOCF is common in clinical settings, it fails to account for the ”smoothness” of mood traits. In the study, spline-based interpolation significantly outperformed LOCF. By contrast, interpolation captures broader trends while reducing the noise associated with short-term volatility. Figure 3 illustrates these methods for two participants with differing recording behaviours. User AS14.33 reported his mood frequently, while user AS14.17 exhibited a gap exceeding our three-day limit. We argue that Linear Interpolation is preferable for our data. As shown in the plot, LOCF produces flat plateaus in the ring gaps. These plateaus can bias machine learning models toward predicting stagnant mood trajectories, an artifact that interpolation avoids by maintaining a trend-based slope.

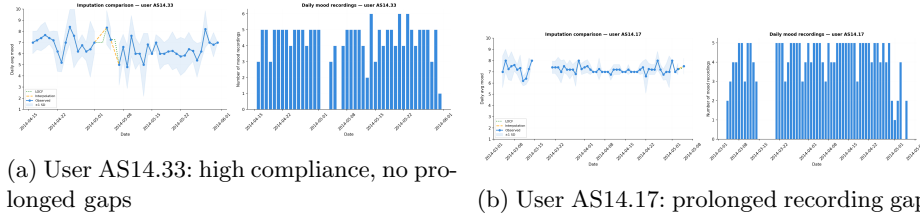


Fig. 3: Comparison of LOCF (green, dotted) and linear interpolation (orange, dashed) for two representative users, where the shaded region denotes ± 1 standard deviation of intra-day mood ratings. The bar chart shows daily mood recording counts, with missing bars indicating days with no self-reports.

2.3 Feature Engineering

The correlation matrix after aggregation and imputation (Fig. 4) indicates that `circumplex.valence.mean` is the only feature with a strong positive linear relationship to `mood.mean`. The weak correlation of others suggests that behavioural impact on mood is likely cumulative or delayed rather than instantaneous. Consequently, to capture the latent temporal dependencies, we moved from a static daily view to a temporal feature engineering. For each participant on prediction day t , we constructed a training instance using features aggregated over a sliding window of the preceding five days, targeting the average mood at day t .

While [3] utilised a 4-day window based on autocorrelation and [5] analysed full weekly cycles, we chose a middle-ground approach (5-day window) to capture sufficient data to model ”mood momentum” while limiting the loss of initial data points for participants with shorter recording spans.

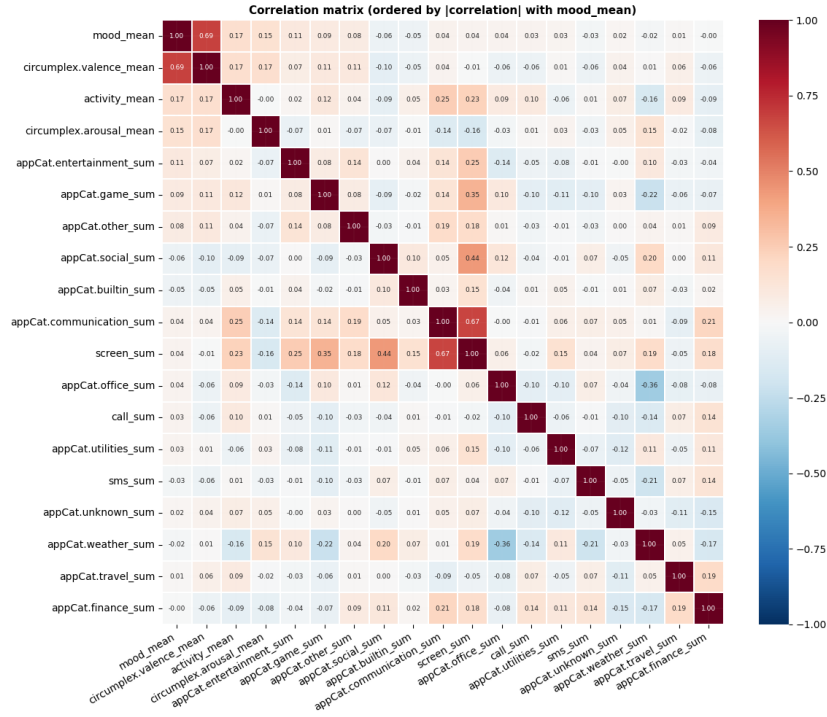


Fig. 4: Correlation matrix

For self-report and activity variables, we calculated standard statistical measures (mean, std, min, max) and two other temporal features: the linear slope (via OLS), to capture the trajectory of the user's state. A positive slope indicates an upward trend, which may be more predictive of next-day mood than a high but declining mean; and the exponential moving average (EMA). Unlike a simple mean, the EMA assigns exponentially decreasing weights to older observations. This reflects the yesterday's mood having typically a stronger influence on today's state compared to a report from four days ago.

For application categories, we calculated absolute usage sums and means. However, absolute duration can be misleading due to high variance in smartphone engagement. Inspired by [2] and [6], we derived normalised usage proportions (histograms). By expressing each category as a percentage of total daily smartphone time, we capture the user's relative allocation of digital attention.

Finally, we encoded day-of-week and month as dummy variables for day t . Day-of-week features were included to capture the influence of social cycles (e.g., weekends versus weekdays), while month variables were retained to serve as proxies for weather and seasonal shifts. This was inspired by [5], who included meteorological conditions to predict new users' well-being. To avoid the dummy variable trap, Sunday and March were omitted as reference categories. The resulting feature space consists of 81 variables across 1,154 training instances.

3 Classification

3.1 Application of Classification Algorithms

As described in Figure 2, daily mood exhibits significant right-skewness and a restricted range. Thus, we discretised the target into three classes, similar to [7]. Given the clinical sensitivity of mood prediction, we considered misclassifying low or high mood days equally costly, which motivated us to prefer a balanced class design. Utilising absolute score boundaries in this distribution would fail to capture psychologically significant shifts. A marginal difference (e.g. 6.8 versus 7.2) effectively represents a total transition from a participant’s relative low to high mood. Therefore, we used quantile-based thresholds ($q_{33} = 6.8$, $q_{66} = 7.25$), which resulted in approximately equal class support.

To ensure a balanced evaluation across mood states, macro-averaged F1 was used for both hyperparameter selection and final evaluation, as it assigns equal weight to each class regardless of the support, directly penalising models that perform well on two classes while neglecting the third, as highlighted by [8]. Additional measures are reported: accuracy for interpretability; Cohen’s kappa κ to measure agreement corrected for chance (providing a more informative baseline comparison than raw accuracy in a three-class setting, [8]); macro-averaged AUC-ROC (one-vs-rest) to assess class discrimination independently of the classification threshold.

Considering the temporal nature of the data, preventing data leakage was critical. Thus, all splits were performed per participant in chronological order, using the `TimeSeriesSplit` function: the first 80% of each participant’s days formed the training set and the final 20% the test set, thus ensuring the model is always evaluated on future observations. For the LSTM, an additional validation set was obtained from the training set using the same temporal logic (70/15/15 split), providing an explicit held-out set for early stopping and hyperparameter selection. Reproducibility was ensured by setting `random_seed`, fixing library versions in `requirements.txt`, using version control and saving models and ensuring a clear pipeline (Figure 1).

3.2 XGBoost Classifier

XGBoost [9] was selected as the first classifier due to its strong empirical performance on tabular data [10], its native robustness to multicollinearity through the greedy splitting mechanism (splits are evaluated independently per feature at each node), and its invariance to feature scaling [11].

All windowed features described in Section 2.3 were used as input. Hyperparameter optimisation was performed using `RandomizedSearchCV` with 50 iterations over a grid covering maximum tree depth [3,6], learning rate [0.01, 0.2], number of estimators [100,500], and row sub-sampling rate [0.8, 1.0], with macro F1 as the scoring criterion (consistent with the primary evaluation metric). The optimal configuration was: `max_depth=5`, `learning_rate=0.01`, `n_estimators=100`, `subsample=0.8`. As the quantile boundaries were very narrow, multi-class log loss was used as XGBoost’s internal evaluation metric.

Table 2 reports main results. 5-fold cross-validation macro F1 (0.520 ± 0.071) and test macro F1 (0.508) are closely aligned, confirming consistent generalisation without overfitting. The test AUC of 0.734 indicates reliable class discrimination above chance, and Cohen’s κ of 0.292 reflects fair agreement beyond a random baseline.

Per-class performance (Figure 5c) reveals a clear asymmetry: low and high mood are predicted substantially better than neutral. We considered this pattern as related to the fact that neutral days occupy the boundary region between the two extreme classes and share behavioural features with both. This possibly makes discrimination harder.

Feature importance (Figure 5b) is dominated by `mood_ema_5d`, the exponential moving average of mood over the past five days, with an importance score more than twice that of the second-ranked feature. Temporal features `weekend`, `Saturday`, `Monday` rank highly alongside total app usage `total_app_time_sum_5d`, suggesting that both behavioural rhythms and usage volume carry predictive signal. Specific app categories such as `entertainment` and `travel` appear in the top 20, though with substantially lower importance than mood history features.

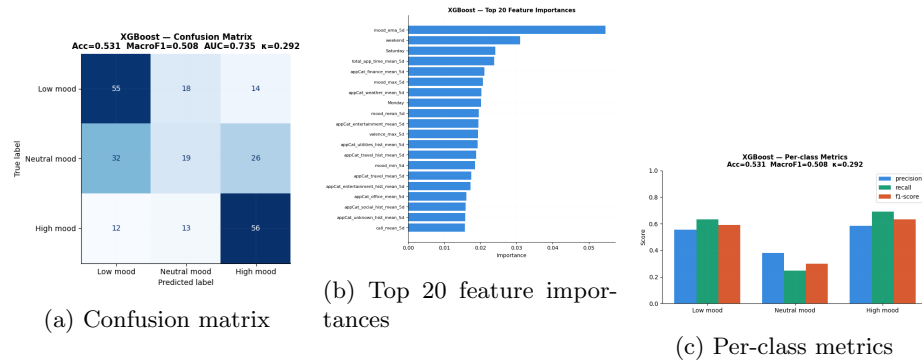


Fig. 5: XGBoost classification evaluation. The confusion matrix (a) shows the model struggles most with Neutral mood. Feature importance (b) is dominated by `mood_ema_5d`. Per-class metrics (c) highlight the asymmetry between neutral mood (F1=0.3), low (F1=0.59), and high mood (F1=0.63) consistent with (a).

LSTM We selected an LSTM network as the second classifier because it is inherently temporal: it processes the raw daily feature sequence directly without requiring window aggregations, allowing the network to learn temporal dependencies from the data itself [12]. The model receives a sequence of seven consecutive days of daily features as input (shape: $7 \times n_{\text{features}}$) and predicts the mood class on the following day. The choice of LSTM (over e.g. Convolutional-Neural Network) and of the 7-days window was also motivated by the mood-related stress prediction study of [13]

Features were normalised using a StandardScaler to prevent features with larger magnitudes from dominating the gradient updates. We opted for an architecture consisting of a single LSTM layer, a dropout layer, a dense layer of

$\max(8, \lfloor \text{units}/2 \rfloor)$ neurons with ReLU activation and L2 regularisation, a second dropout layer, and a three-way softmax output. We selected a compact architecture with strong regularisation (dropout and L2) to ensure overfitting was mitigated, given the limited dataset size. This was verified through the training curves in Figure 7.

A grid search was performed over 32 hyperparameter combinations covering LSTM units (16, 32), dropout rate (0.25, 0.40), learning rate (5×10^{-4} , 1×10^{-3}), batch size (16, 32), and L2 regularisation strength (0.001, 0.01) to select the best model based on macro F1. Models were trained for up to 80 epochs with early stopping, monitoring validation loss and restoring best weights. The absolute `with_sum` and normalised `with_hist` feature sets for app categories described in Section 2.3 were evaluated separately, since, unlike XGBoost, feeding both absolute and proportional features to LSTM would introduce redundancy in the gradient update.

The optimal configuration for `with_hist` was: `units=32`, `dropout=0.25`, `lr=0.0005`, `batch_size=32`, `l2_reg=0.01`, converging at epoch 76. Validation and test set results for both feature conditions are reported in Table 2. Overall, test and validation macro F1 measures confirm generalisation without overfitting. The test AUC indicates comparable class discrimination to XGBoost, and Cohen’s κ reflects moderate agreement beyond chance.

The `with_hist` feature set outperformed `with_sum` on accuracy and macro F1, suggesting that normalised app usage proportions provide a more informative representation than absolute durations, as they control for day-to-day variation in total phone usage. The training curves in (Figure 7) support `with_hist` better performance over `with_sum`. While the first one shows a negative train-val gap, indicating no overfitting, the latter exhibits a small positive gap, suggesting the raw duration features introduce mild overfitting despite regularisation.

Both feature sets reproduce the per-class asymmetry observed with XGBoost (Figure 8): neutral mood remains the hardest class to predict, nonetheless, improvement on XGBoost results.

Table 2: Classification performance. Validation (Val.) macro F1 is mean $\pm$ std across five temporal folds for XGBoost. Best validation macro F1 for LSTM.

	Val. F1	Acc	Macro F1	AUC	κ
XGBoost	0.520 ± 0.071	0.531	0.508	0.734	0.292
LSTM (<code>with_hist</code>)	0.622	0.554	0.556	0.729	0.333
LSTM (<code>with_sum</code>)	0.627	0.491	0.488	0.700	0.236

3.3 Winning Classification Algorithms

We chose to look at a Kaggle competition that closely relates to the concepts explored in our dataset. The Child Mind Institute held the competition in the fall of 2024, with the deadline set in December and participating teams had 3

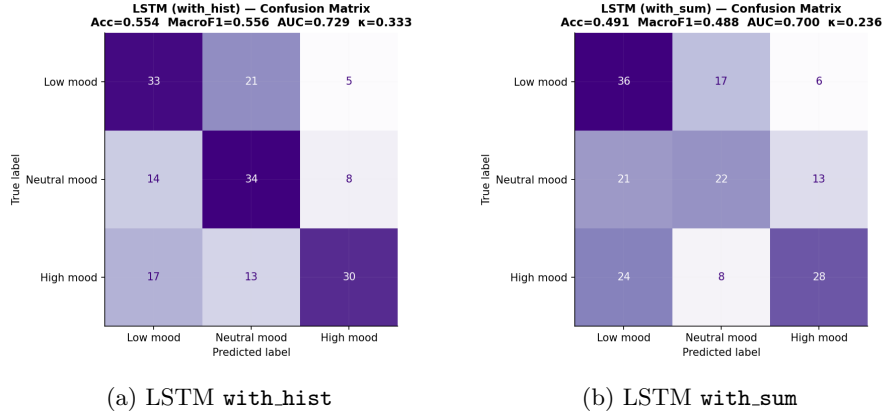


Fig. 6: LSTM confusion matrices for both feature sets. Both conditions reproduce the Neutral mood weakness observed with XGBoost, though the `with_hist` set reduces Neutral mood misclassification compared to `with_sum` (34 correct vs 22 correct).

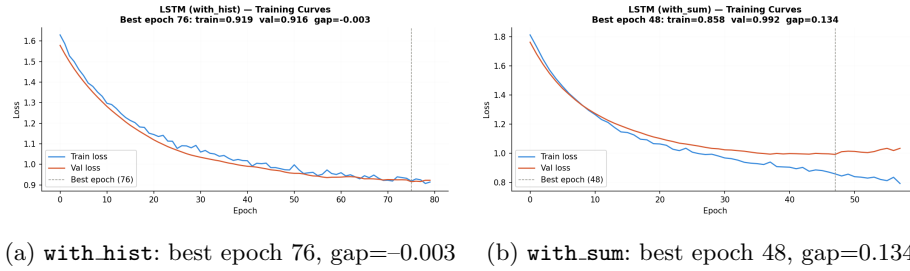


Fig. 7: LSTM training curves for both feature sets. Early stopping restores weights at the best validation epoch (dashed line).

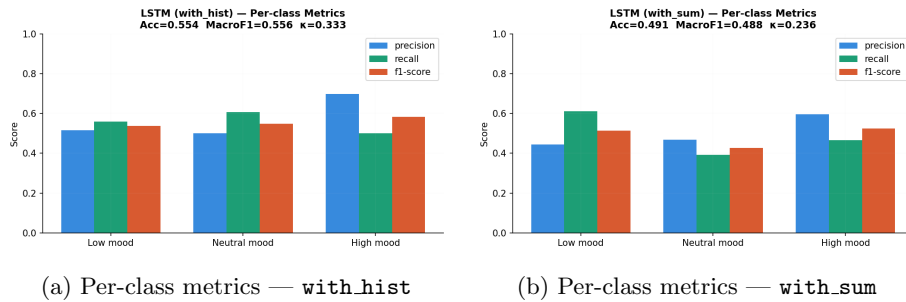


Fig. 8: Per-class precision, recall and F1 for the two LSTM configurations. Neutral mood consistently achieves the lowest F1 across both sets.

months to develop their solutions. The solution write-ups after the victors were chosen were published in January of 2025. The competitors were tasked with predicting the severity of problematic internet use in children and young adults. The data consisted of 5,000 participants between the ages of 5 and 22 collected by the Healthy Brain Network (HBN) [14]. The Kaggle competitors mainly used two aspects of the original HBN dataset, the physical activity data (wrist-worn accelerometer data, fitness assessments and questionnaires) and internet usage behaviour data. The severity of problematic internet use was measured by the Severity Impairment Index (SII) - an ordinal variable with four levels ranging from 0 (no impairment) to 3 (severe impairment) [14] which is derived from participants Parent Child Internet Addiction Test (PCIAT) [15]. The competition was evaluated using the Quadratic Weighted Kappa (QWK), which is used to measure the agreement between predicted and true ordinal labels [16]. The QWK formula is detailed as:

$$\kappa = 1 - \frac{\sum_{i,j} w_{ij} O_{ij}}{\sum_{i,j} w_{ij} E_{ij}}$$

where O is the matrix constructed from the number of samples with actual value and the number of samples with a predicted value, E is the matrix of expected outcomes and w is used to penalise misclassifications based on the distance of error [16].

Lennart Haupt won the competition by reframing the problem as a regression task focused on predicting the underlying continuous PCIAT total score, which was later converted into ordinal classes using optimised thresholds. His final technique consisted of a majority-vote ensemble of five techniques: LightGBM, two XGBoost variants, CatBoost and ExtraTreesRegressor [17]. Each model individually obtained its own predictors, which were then converted to ordinal classes via the thresholds. Additionally, he employed Lasso imputation for features with fewer than 40 missing values, defaulting to mean imputation only when the Lasso model could not be trained [17]. The main driver of his success was that he first predicted the PCIAT score, from which the SII score is derived from using optimized thresholds. Haupt chose not to fix these thresholds and instead optimized within each cross-validation folds to maximise his QWK score. Haupt demonstrated that the QWK is highly sensitive to class boundaries, by optimising the class boundaries directly against the QWK within each fold, he tuned the model to perform well on the exact metric used to rank him [17].

A possible standard approach to this problem would be to train a multiclass classifier that would directly predict one of the four SII ordinal levels. In this approach all wrong predictions are treated equally, classifying a level 3 child as 0 or 2 is treated as the same mistake. This is not the case for the QWK evaluation metric [16], which penalizes farther predictions more severely than closer predictions when both are wrong. Haupt adapted to this by predicting his PCIAT scores using a regression and then converting those predictions to the ordinal level thresholds. Additionally, he ensured the thresholds were not fixed and optimized each cross-validation fold, thereby reducing his chance of mistaken

classifications. Finally, his use of a five vote ensemble of models separated him from his competitors. By using five different regressors, he reduced his exposed risk to a single model's errors dominating the final prediction [18].

4 Association Rules

The APRIORI algorithm introduced by Srikant and Agrawal [19], aims to identify frequent item sets and generate association rules between items, such that "customers who buy item x also buy item y ". A limitation of this algorithm is the computational cost: scanning the entire database, once for each itemset size. Increasing item stocks size leads to considerable process deceleration. Tanian et al. [20], build upon the APRIORI algorithm, looking at mining association rules across multiple levels of a preset hierarchy and measuring negative associations. The "Hierarchy Top-Down Mining Algorithm", implemented by Tanian et al [20] begins at the highest and broadest level of the predefined hierarchy (e.g. "Food") and searches for categories that are often bought together. The categories are broad, thus links are easy to achieve. It then repeats that process with lower-level categories, always searching for ones that are frequently bought together. At the lowest level, only then does the algorithm search for items that are frequently bought together. During each level search, the algorithm searches for items or categories that are rarely bought in tandem. Each level search has its own support thresholds, where lower, more specific levels have more relaxed thresholds than higher, broader levels [20].

A strength of Tanian et al. [20] model is that it allows for early pruning of branches, thus increasing computational efficiency and addressing one of the limitations of APRIORI. Another strength lies in its measurement of negative associations, thereby increasing the richness of the results. However, the algorithm's performance is limited by the accuracy and relevance of its predefined hierarchical framework. Unlike the original APRIORI algorithm, which requires no prior structure, this model necessitates manual, intelligent design based on domain expertise. Another limitation lies in the need for support thresholds at every level, which requires additional parameter turning by experts.

5 Numerical Prediction

5.1 XGBoost Regressor

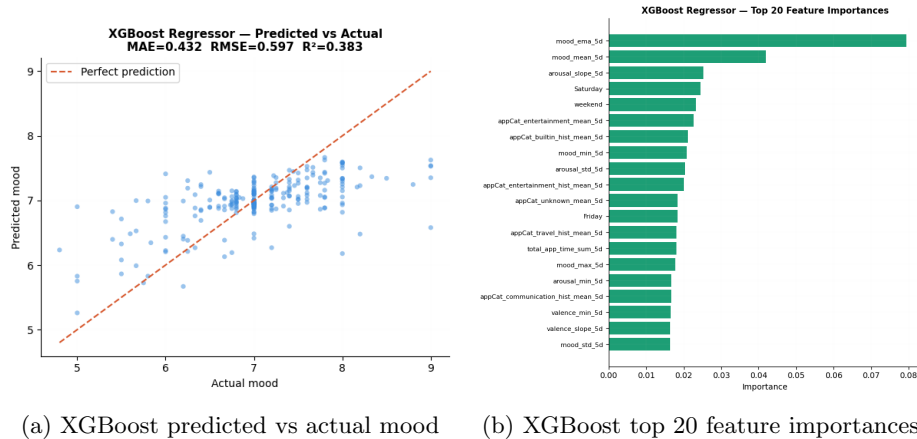
We decided to apply the same tabular feature set and temporal evaluation setup described in Section 3 to the regression task, with the continuous daily average mood as the target, given XGBoost robustness properties mentioned in Section 3.2.

Similarly to the classification problem, the model was optimised for **squarederror** using **RandomizedSearchCV** with 50 iterations over the same grid as the classifier, using the group-aware temporal CV and Root Mean Squared Error (RMSE) as

the scoring criterion. The optimal configuration was: `max_depth=3`, `learning_rate=0.01`, `n_estimators=500`, `subsample=0.8`.

The model demonstrated stability and a clear predictive signal. On the held-out test set the model achieved $\text{MAE}=0.432$, $\text{RMSE}=0.597$, and $R^2=0.383$, indicating that it explains approximately 38% of the variance in next-day mood. The close agreement between CV and test RMSE shown in Table 3 confirms consistent generalisation without overfitting. An analysis of the MSE/MAE ratio (0.826) suggests that the model’s error is composed of consistent, minor residuals rather than large outliers.

In alignment with established psychological literature, Figure 9b mirrors classification findings, that past mood state is the primary predictor of future mood. Emotional arousal is also predictive, as well as temporal context, with Saturday and weekend ranking as top-five predictors, suggesting that weekly behavioural rhythms are more influential than specific app-usage categories.



(a) XGBoost predicted vs actual mood (b) XGBoost top 20 feature importances

Fig. 9: XGBoost regressor evaluation. Predicted vs actual (a) shows a positive correlation with the diagonal, with most predictions clustered around the mean mood range (6–8). Feature importance (b) is again dominated by `mood_ema_5d` and `arousal_slope_5d`.

5.2 LSTM Regressor

We used LSTM network also as a temporal approach in numerical prediction, given the theoretical capacity of such model the non-linear, long-range dependencies inherent in mood fluctuations. The LSTM regression model shares the architecture of the classification LSTM (mitigating overfitting), with two modifications: the output layer is a single neuron with linear activation, and the training loss is MSE. The same grid of 32 hyperparameter combinations was evaluated, with model selection based on validation RMSE, consistent with the primary metric. We used the 70/15/15 temporal split (Train/Validation/Test)

per participant, where the validation set served for early stopping. The optimal configuration for both feature sets was identical: `units=32`, `dropout=0.25`, `lr=0.001`, `l2_reg=0.01` with only `batch_size=32` with `hist` and 16 or with `sum`.

Test set results are reported in Table 3. Both LSTM configurations substantially underperformed the XGBoost baseline, with higher errors and negative R^2 . A negative R^2 indicates that both models perform worse than the trivial baseline of always predicting the mean mood, which is a notable failure.

This is consistent with the known difficulty of training recurrent networks on small datasets [10]: the dataset contains only 27 participants, providing limited diverse sequences for the LSTM to learn stable temporal representations. XGBoost, on the other hand, is less sensitive to dataset size. Our results shows that the optimal choice of algorithm might depend on the target we set: class or numerical mood.

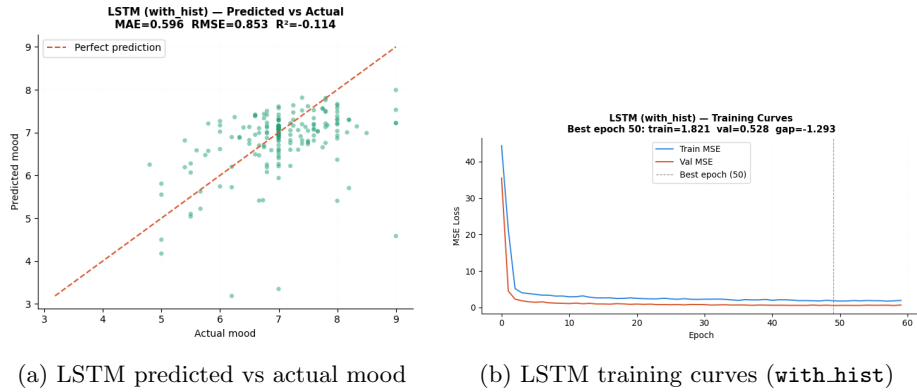


Fig. 10: LSTM regressor evaluation (`with_hist`). (a) Predictions cluster around the mean mood range with limited spread, reflecting the negative R^2 . (b) Training curves show a negative train-val gap, which could still be related to temporal per-patient splitting on small datasets rather than true generalization.

Table 3: Regression performance. Validation (Val.) RMSE is mean $\pm$ std across five temporal folds for XGBoost; best validation RMSE for LSTM.

	Val. RMSE	MAE	RMSE	R^2	MSE/MAE
XGBoost	0.614 ± 0.041	0.432	0.597	0.383	0.826
LSTM (<code>with_hist</code>)	0.853	0.622	0.853	-0.114	1.222
LSTM (<code>with_sum</code>)	0.786	0.638	0.786	-0.054	1.098

6 Evaluation Metrics

6.1 Characteristics of MSE and MAE

Mean Squared Error (MSE) and Mean Absolute Error (MAE) are defined as:

$MSE = \frac{1}{n} \sum_i (y_i - \hat{y}_i)^2$ $MAE = \frac{1}{n} \sum_i |y_i - \hat{y}_i|$ The key difference is the squaring operation in MSE: an error of magnitude ϵ contributes ϵ^2 to MSE but only ϵ to MAE. For errors greater than 1, squaring amplifies larger errors disproportionately, making MSE more sensitive to outlier predictions. For errors less than 1, squaring reduces the contribution, making MSE less sensitive to very small errors than MAE. Consequently, MSE is preferred when large errors are disproportionately costly (e.g. safety-critical applications), while MAE is preferred when all errors should be penalised proportionally and robustness to outliers is desired. Furthermore, MSE and MAE yield identical values when all residuals have the same absolute magnitude. Formally, let $|\epsilon_i| = c$ for all i . Then $MAE = c$ and $MSE = c^2$, so $MSE = MAE$ only when $c = 1$. More generally, $MSE = MAE$ when the variance of squared residuals is zero, i.e. all prediction errors are exactly equal in absolute value.

6.2 Impact of Evaluation Metrics on Regression Results

Table 3 reports both MAE and RMSE ($\sqrt{MSE}$) for all regression models. For the best-performing LSTM model `with_sum`, the MSE/MAE ratio is 1.098, which is close to 1.0. It indicates that the prediction errors are relatively uniform in magnitude and that extreme outliers do not inflate it greatly. This near-unity ratio indicates that prediction errors are highly uniform in magnitude, no extreme outlier predictions inflate MSE disproportionately. This is consistent with the narrow distribution of daily average mood ($\text{std} \approx 0.7$ after daily averaging), which usually limits the range of residuals. In this context, MSE and MAE lead to essentially the same conclusions about model performance.

For XGBoost, the MSE/MAE ratio is 0.826. It provides context to the flattened distribution seen in Figure 9. It suggests that most of the errors are small, with very few predictions having errors larger than one full point on the mood scale. As the model predicts near the mean for all instances, the resulting errors are compressed by the squaring operation in MSE, leading to a ratio below 1.0. Given the narrow target distribution, the absence of extreme outlier errors and the subjective nature of self-reporting the mood, MAE is the more appropriate primary metric for this task, as it is interpretable in mood scale units and does not overweight the occasional difficult prediction day.

7 Conclusion

This report applied the CRISP-DM process to predict next-day mood from smartphone sensor data. Following temporal cleaning, gap handling, and rolling window feature engineering, XGBoost consistently outperformed the LSTM for both classification and regression, consistent with the known data efficiency advantage of tree-based models on small datasets.

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